

# DATA266 Lab1 Team Report

Team number: 31 | Submission date: 10/06/2026

Member A: Omkar Rajale | Member B: Sai Sushma Maddali

Shared GitHub repository: <https://github.com/sai-sushma-maddali/data266-lab1/tree/main>

## Team ownership statement

Omkar Rajale independently designed, trained, and evaluated his own models for Tasks 1, 2, and 3 (GPT-style char-LM v4 wider Transformer; FastText baseline + TextCNN + Transformer sentiment models; CycleGAN ResNet-9 with DiffAugment + generator EMA). Sai Sushma Maddali independently designed, trained, and evaluated her own models for Tasks 1, 2, and 3 (GPT-style char-LM v8; Baseline GRU + Bidirectional LSTM + Attention LSTM sentiment models; CycleGAN with DiffAugment, EMA, and feature-matching). Together we assembled this combined report, compared architectures and metrics side by side, and aligned evaluation settings where possible (official CycleGAN FID/MiFID protocol with N=300).

## Report notes

This report replaces the template placeholders with measured results from each member's notebooks. Where a metric was not computed or not saved in notebook outputs, we explicitly mark it as not measured.

## Shared evaluation and reproducibility

| Item                     | Details                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Task evaluation scripts  | Task 1: each member's notebook metric cells (CE, perplexity, BPC, Distinct-n, etc.). Task 2: sklearn + bootstrap + McNemar in each member's training notebook. Task 3: Omkar Part3_Evaluation_Script_v7; Sushma Task3_CycleGAN_Official_Evaluation-v5 (official FID/MiFID, N=300).                                                                                                                              |
| Data and splits          | Task 1: TinyStories; each member used own 100K train / 10K val split (Omkar seed 298; Sushma seed 5775). Char tokenization with independent vocabs (Omkar vocab 114 after cleaning; Sushma vocab 152). Task 2: Yelp Polarity; Omkar used HF train/test with own vocab (30K, max_len 256); Sushma used 504K/56K/38K train/val/test split, vocab 40K, seed 5775. Task 3: Monet↔Photo unpaired domains at 256×256. |
| Hardware and environment | Omkar: NVIDIA GeForce RTX 5090 (~32 GB) for Tasks 1–2; NVIDIA A100-SXM4-80GB for Task 3 (torch 2.7.1+cu128 / 2.11.0+cu130). Sushma: NVIDIA GeForce RTX 5090 for Task 1; NVIDIA L4 (22 GB) for Task 2; Task 3 training/eval on lab GPU environment (seed 5775).                                                                                                                                                  |
| Checkpoint mapping       | Task 1 Omkar: version_4 best_model.pt (epoch 10). Task 1 Sushma: best model at epoch 15 (val loss 0.5242). Task 2 Omkar: FastText_Baseline.pt, TextCNN_Exp1, Transformer_Exp2. Task 2 Sushma: baseline_gru, bidirectional_lstm, attention_lstm best_model.pt. Task 3 Omkar: best.pt epoch 40 EMA (cyclegan_r9_diffaug_ema). Task 3 Sushma: best_cycle_checkpoint / diffaug_ema generators (see Task 3 table).   |
| Reproduction command     | See shared repo README for the documented one-command smoke test per task folder (task1_llm/, task2_sentiment/, task3_gan/). Notebooks in Omkar-code/ and Sushma-code/ are the primary runnable sources for this draft.                                                                                                                                                                                         |

Final PDF location: report/DATA266\_Lab1\_Report\_Team\_31.pdf

# Task 1 GPT style LLM from scratch

Dataset: TinyStories. Each member used character tokenization and an independent 100K training / 10K validation split. Omkar's reported run is Version 4 (wider Transformer). Sushma's reported run is Task\_1 v8.

| Item                                    | Omkar Rajale                                                                                           | Sai Sushma Maddali                                                                       |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Model and run ID                        | GPT-char LM version_4 (wider_transformer), run version_4_20260918T201955Z                              | GPT-char LM v8, seed 5775, HP_ID 3                                                       |
| Architecture and parameter count        | Decoder-only Transformer from scratch; 25,582,592 params                                               | Decoder-only Transformer from scratch; 25,506,968 params                                 |
| Layers / heads / embedding size         | 8 layers / 8 heads / emb 512 / FFN 2048 / dropout 0.1                                                  | 8 layers / 8 heads / hidden 512 / FFN 2048 / dropout 0.1                                 |
| Context length / vocabulary size        | block_size 512 / vocab 114                                                                             | sequence_length 256 / vocab 152                                                          |
| Optimizer / learning rate / batch size  | AdamW; max LR 3e-4 (min 3e-5); WD 0.1; physical batch 16 x grad_accum 2 (effective 32)                 | Adam fused; LR 1e-4; batch 64; no WD printed                                             |
| Warm up / scheduler / epochs / seed     | Linear warm-up 5% steps + cosine; 10 epochs; seed 298; max grad norm 1.0; BF16                         | Linear warm-up (1000 steps) + cosine decay; 15 epochs; seed 5775                         |
| Training cross entropy loss             | 0.473963 (best-epoch train)                                                                            | 0.4850 (best-model train eval)                                                           |
| Validation cross entropy loss           | 0.481881                                                                                               | 0.5242                                                                                   |
| Perplexity                              | 1.619117 (from val CE)                                                                                 | 1.6891 (val)                                                                             |
| Bits per character                      | 0.695207                                                                                               | 0.7563 (val)                                                                             |
| Generalization gap and definition       | 0.007918 (val – train CE at best epoch)                                                                | 0.0392 (val – train CE)                                                                  |
| Top 1 next character accuracy           | 84.5072% (val)                                                                                         | 83.1884% (val); train 84.2200%                                                           |
| Distinct 1 / Distinct 2 / Distinct 3    | 0.017971 / 0.135437 / 0.354117                                                                         | 0.0115 / 0.0897 / 0.2548                                                                 |
| Repeated 4 gram rate                    | 0.457017                                                                                               | 0.5865                                                                                   |
| Gradient norm / loss spikes / NaN count | Stable; NaN/non-finite 0; per-step grads logged (late ~0.23-0.45); run avg/max NOT aggregated in print | Avg grad norm 0.3377; max 4.5985; NaN/Inf 0                                              |
| Training tokens per second              | Overall 250,723 tokens/sec (epoch logs ~245k-254k)                                                     | 287,496.25 tokens/sec                                                                    |
| Generation tokens per second            | Avg ~340.82 (metric pass); sample gens ~180–287 tok/s                                                  | 215.92                                                                                   |
| Peak memory and total training time     | Peak allocated GPU 3.60 GB; total training 3736.74 s (~62.3 min)                                       | Peak GPU 12,244.10 MB (~12.0 GB); training-only 9362.95 s (~2.60 h); train+val 9615.01 s |
| Decoding settings and sample count      | Standard metrics: T=0.8, top-k=40; also quality decode T=0.7 top-p=0.9 + newline stop (not retrained)  | Temperature T=0.4; 10 prompts x 2 samples; 200 new chars each                            |
| Metric CSV / log / checkpoint links     | task1_artifacts/version_4/ (logs, checkpoints, generated_samplesv4_standard.txt)                       | Task_1 v8 notebook outputs + saved Drive artifacts                                       |

Training/validation loss curves are embedded below (Omkar: task1\_llm/Omkar/outputs/training\_metrics\_v4.png; Sushma: task1\_llm/Sushma/outputs/loss\_curves\_v8.png).

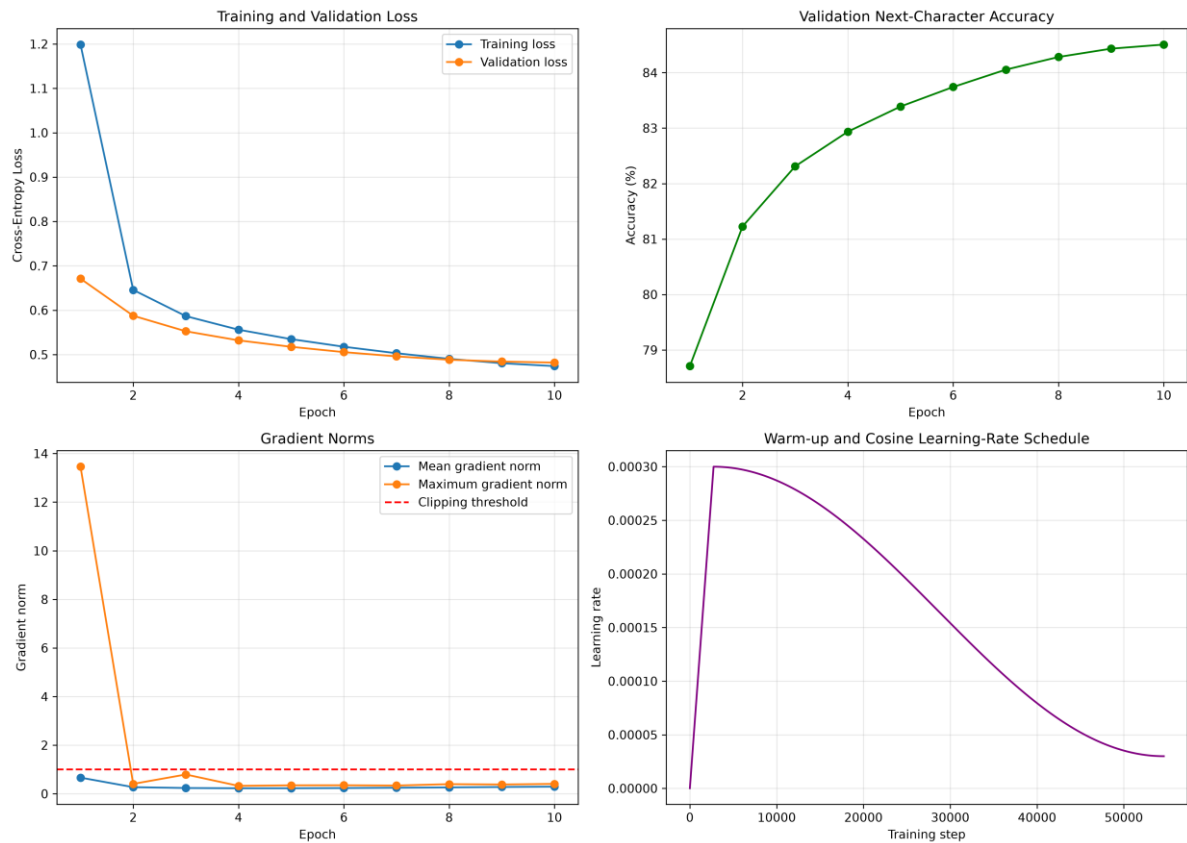


Figure T1-1. Omkar (v4) training metrics / loss curves (task1\_llm/Omkar/outputs/training\_metrics\_v4.png).

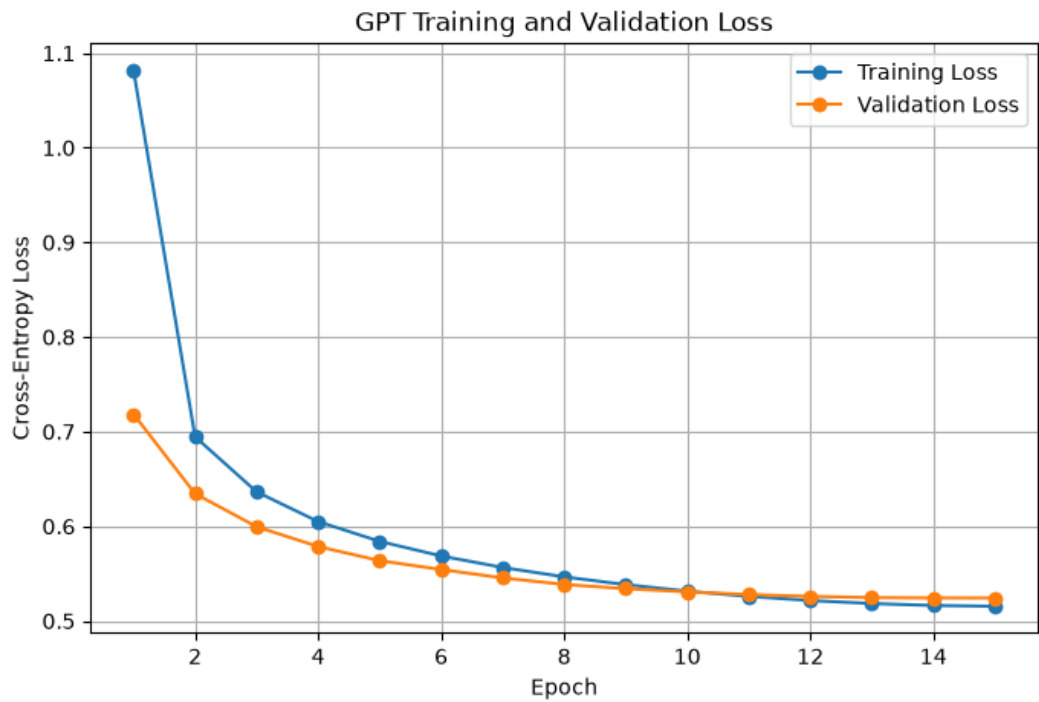


Figure T1-2. Sushma (v8) training and validation cross-entropy (task1\_llm/Sushma/outputs/loss\_curves\_v8.png).

# Task 1 Joint analysis and failure cases

## Strengths

Strengths: Both of us successfully trained from-scratch causal Transformers on TinyStories without prebuilt attention modules. Omkar's wider 512-context model reached lower validation CE (0.482) and higher top-1 accuracy (84.51%) with a very small generalization gap (0.0079). Sushma's longer 15-epoch schedule also converged smoothly (NaN count 0) and produced readable short stories under temperature sampling. Both runs remained numerically stable with warm-up + cosine schedules and BF16.

## Weaknesses and limitations

Weaknesses and limitations: Character-level modeling with ~25M parameters still struggles with long-range narrative coherence. Generation diversity metrics are low (Distinct-1 < 0.02) and repeated 4-gram rates are high (Omkar 0.46; Sushma 0.59), matching qualitative repetition and entity-tracking failures. Vocab and cleaning differences (114 vs 152) plus different context lengths (512 vs 256) and seeds mean absolute metric comparisons are informative but not perfectly controlled.

## Team comparison

Team comparison: Architectures are similar in depth/width (8×8×512), but Omkar used AdamW with higher LR (3e-4), longer context (512), gradient accumulation, and a cleaned smaller vocab; Sushma used Adam at 1e-4, context 256, larger vocab, and 15 epochs. Omkar's best val CE/perplexity are stronger; Sushma trained longer and reported higher peak memory. Neither model fully solves story-level coherence despite strong next-char accuracy.

## What we would try next

What we would try next: (1) Add a small BPE/subword vocabulary or hierarchical decoding to reduce character-level repetition while keeping the from-scratch constraint spirit for analysis. (2) Train with stronger generation-aware regularization (repetition penalty during sampling, or unlikelihood on repeated n-grams) and evaluate Distinct-n / repeated-4-gram as first-class training monitors.

## Omkar Rajale — Three failure cases

Case 1: Prompt "The little dog was afraid" -> story continues then abruptly restarts: "From that day on, the little dog helped the kind man whenever he collected flowers." then "Once upon a time, there was a little girl named Lily..."

Failure type: Loss of coherence / abrupt second-story restart

Observation and source: Standard V4 decoding (T=0.8, top-k=40); generated\_samples under task1\_artifacts/version\_4/outputs/.

Case 2: Barrel scene - "Lily wanted to sit on it. She ran to the barrel and pulled and tugged until it broke. The barrel was shut and smashed." ... trailing broken token "Ben he"

Failure type: Broken grammar / incoherent object behavior

Observation and source: Same V4 standard generations; objects act inconsistently and text truncates mid-token.

Case 3: Quality decoding (T=0.7, top-p=0.9) still yields odd agency: "In a small forest, there was a big tree... One day, the tree was hungry."

Failure type: Unsupported / illogical details

Observation and source: Quality-focused decode (no retrain); readability improved but pronoun/object issues remain.

## Sai Sushma Maddali — Three failure cases

Case 1: Bow repetition loop - "She opened the box and saw a big bow. It was a shiny red bow." ... "She put the bow in the bow. She said, \"This is a special bow. It is a special bow. It is a special bow. It is a special bow.\""

Failure type: Repetition

Observation and source: Task\_1 v8 generation samples; matches high repeated-4-gram rate (0.5865).

Case 2: Low-diversity temperature sample (T=0.2) - Nora finds a tiny door, then suddenly a crying girl appears with weak causal link.

Failure type: Loss of coherence / abrupt character insertion

Observation and source: Temperature sweep excerpt in Task\_1 v8.

Case 3: Attic/bear bakery sample ends mid-sentence ("Then they both started t") after an unmotivated topic jump from fear to pastry.

Failure type: Truncation / topic shift

Observation and source: T=0.2 generation in Task\_1 v8 cell output.

## Task 2 Yelp Polarity sentiment classification

One baseline and two experimental models per member. All embeddings learned from scratch. Omkar: FastText baseline, TextCNN, Transformer encoder classifier. Sushma: Baseline GRU, Bidirectional LSTM, Attention LSTM. Note: splits differ (Omkar HF test set vs Sushma 38K held-out test), so cross-member accuracy is not an identical protocol comparison.

| Item                                     | Omkar Baseline                                                        | Omkar Experiment 1                                           | Omkar Experiment 2                                           | Sushma / Baseline GRU                                                   | Sushma / BiLSTM Exp1                                     | Sushma / Attn-LSTM Exp2                           |
|------------------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------------------|----------------------------------------------------------|---------------------------------------------------|
| Model / run ID                           | FastText_Baseline                                                     | TextCNN_Exp1                                                 | Transformer_Exp2                                             | baseline_gru                                                            | bidirectional_lstm                                       | attention_lstm                                    |
| Architecture and embedding justification | Mean-pool word embeddings + linear (FastText-style bag-of-embeddings) | Multi-filter TextCNN (filters 3/4/5) over learned embeddings | Small Transformer encoder + positional encoding + classifier | Unidirectional GRU over learned embeddings; last hidden → logit         | Bidirectional LSTM; concat final forward/backward states | LSTM + additive attention pooling over time steps |
| Embedding size / vocabulary / max length | 128 / 30K / 256                                                       | 128 / 30K / 256                                              | 128 / 30K / 256                                              | 128 / 40K / max_len 256                                                 | 128 / 40K / max_len 256                                  | 128 / 40K / max_len 256                           |
| Optimizer / learning rate / batch size   | AdamW / 1e-3 / 512                                                    | AdamW / 1e-3 / 512                                           | AdamW / 5e-4 / 512                                           | AdamW / 1e-3 / train batch 256 (eval batch 512)                         | AdamW / 1e-3 / train batch 256                           | AdamW / 1e-3 / train batch 256                    |
| Epochs / dropout / seed / other changes  | 5 epochs; seed from Omkar run setup                                   | 5 epochs; dropout 0.5 in CNN                                 | 5 epochs; lower LR vs baseline                               | 6 epochs; dropout 0.30; seed 5775; ReduceLROnPlateau (min_lr 1e-5); AMP | 6 epochs; dropout 0.30; bidirectional; same schedule     | 6 epochs; dropout 0.30; attention; same schedule  |
| CPU and GPU hardware                     | NVIDIA RTX 5090                                                       | NVIDIA RTX 5090                                              | NVIDIA RTX 5090                                              | NVIDIA L4 22 GB                                                         | NVIDIA L4 22 GB                                          | NVIDIA L4 22 GB                                   |
| Split sizes / preprocessing / threshold  | Yelp polarity HF train+test; lowercasing/clean; threshold 0.5         | Same as Omkar baseline                                       | Same as Omkar baseline                                       | 504K/56K/38K; cleaning+encode; threshold 0.5                            | Same as Sushma baseline                                  | Same as Sushma baseline                           |
| Accuracy                                 | 0.9355                                                                | 0.9433                                                       | 0.9302                                                       | 0.9517                                                                  | 0.9533                                                   | 0.9506                                            |
| Precision macro / micro / weighted       | ≈0.9355 (macro F1 matches; micro≈acc)                                 | ≈0.9433                                                      | ≈0.9302                                                      | 0.9518 / 0.9517 / 0.9518                                                | 0.9533 / 0.9533 / 0.9533                                 | 0.9506 / 0.9506 / 0.9506                          |
| Recall macro / micro / weighted          | ≈0.9355                                                               | ≈0.9433                                                      | ≈0.9302                                                      | 0.9517 / 0.9517 / 0.9517                                                | 0.9533 / 0.9533 / 0.9533                                 | 0.9506 / 0.9506 / 0.9506                          |
| F1 macro / micro / weighted              | 0.9355 / ≈0.9355 / ≈0.9355                                            | 0.9433 / ≈0.9433 / ≈0.9433                                   | 0.9302 / ≈0.9302 / ≈0.9302                                   | 0.9517 / 0.9517 / 0.9517                                                | 0.9533 / 0.9533 / 0.9533                                 | 0.9506 / 0.9506 / 0.9506                          |
| Confusion matrix and evidence link       | Computed in evaluate(); see notebook outputs                          | Computed in evaluate(); TextCNN used for error review        | Computed in evaluate()                                       | [[17964,1036],[798,18202]]                                              | [[18102,898],[876,18124]]                                | [[18059,941],[938,18062]]                         |
| ROC AUC                                  | 0.9790                                                                | 0.9862                                                       | 0.9836                                                       | 0.9905                                                                  | 0.9896                                                   | 0.9895                                            |
| PR AUC                                   | 0.9785                                                                | 0.9867                                                       | 0.9842                                                       | 0.9907                                                                  | 0.9897                                                   | 0.9897                                            |
| Matthews correlation coefficient         | 0.8710                                                                | 0.8867                                                       | 0.8618                                                       | 0.9035                                                                  | 0.9066                                                   | 0.9011                                            |
| Brier score                              | 0.0514                                                                | 0.0429                                                       | 0.0519                                                       | 0.0362                                                                  | 0.0370                                                   | 0.0380                                            |
| Expected calibration error               | 0.0166                                                                | 0.0048                                                       | 0.0306                                                       | 0.0121                                                                  | 0.0195                                                   | 0.0170                                            |

| Item                                    | Omkar Baseline                                                            | Omkar Experiment 1                                                     | Omkar Experiment 2                                                     | Sushma / Baseline GRU                                                   | Sushma / BiLSTM Exp1                                | Sushma / Attn-LSTM Exp2                                        |
|-----------------------------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------|-----------------------------------------------------|----------------------------------------------------------------|
| Accuracy 95% bootstrap CI               | Not printed in saved notebook output (bootstrap fn present; n=300)        | Not printed in saved notebook output                                   | Not printed in saved notebook output                                   | [0.9496, 0.9539] (2000 bootstraps)                                      | [0.9512, 0.9555]                                    | [0.9482, 0.9526]                                               |
| Macro F1 95% bootstrap CI               | Not printed in saved notebook output                                      | Not printed in saved notebook output                                   | Not printed in saved notebook output                                   | [0.9496, 0.9539]                                                        | [0.9512, 0.9555]                                    | [0.9482, 0.9526]                                               |
| MCC 95% bootstrap CI                    | Not printed in saved notebook output                                      | Not printed in saved notebook output                                   | Not printed in saved notebook output                                   | [0.8993, 0.9078]                                                        | [0.9023, 0.9111]                                    | [0.8965, 0.9053]                                               |
| McNemar vs baseline statistic / p value | N/A                                                                       | Computed vs FastText; saved to manifest (values not printed in output) | Computed vs FastText; saved to manifest (values not printed in output) | N/A                                                                     | exact p=0.0620 vs GRU (Holm 0.124; not sig at 0.05) | exact p=0.1848 vs GRU (not sig); BiLSTM vs Attn p=0.0020 (sig) |
| Slice names / macro F1 / error rate     | Long-review slice used in error analysis (>200 words)                     | Same; TextCNN long-slice failures reviewed                             | Same protocol available                                                | short/med/long + negation + truncation; e.g. short F1 0.9511 err 0.0475 | Best on most slices (e.g. long 0.9507 acc)          | Weaker on truncated slice (0.9293 acc)                         |
| Parameter count                         | 3,840,129                                                                 | 3,994,201                                                              | 4,113,281                                                              | 5,219,201                                                               | 5,384,449                                           | 5,252,354                                                      |
| Training time                           | 127.70 s                                                                  | 120.77 s                                                               | 123.30 s                                                               | 271.40 s                                                                | 406.54 s                                            | 466.92 s                                                       |
| Examples per second and definition      | 21,925.6 train ex/s (dataset×epochs / wall)                               | 23,185.4                                                               | 22,708.0                                                               | Train 11,700 ex/s; infer 21,242 ex/s                                    | Train 7,685; infer 24,033                           | Train 6,665; infer 21,885                                      |
| Peak memory                             | 0.161 GB                                                                  | 0.408 GB                                                               | 1.507 GB                                                               | Train peak 0.182 GB                                                     | 0.282 GB                                            | 0.271 GB                                                       |
| Metric CSV / log / checkpoint links     | Omkar Task-2 notebook + outputs/model_comparison_table.csv + checkpoints/ | Same                                                                   | Same                                                                   | Sushma Task_2 notebook outputs + Drive checkpoints/baseline_gru/        | checkpoints/bidirectional_lstm/                     | checkpoints/attention_lstm/                                    |

## Task 2 Joint analysis

### Strengths

Strengths: All six models beat chance by a wide margin with learned-from-scratch embeddings only. Omkar's TextCNN was his best model (acc 0.9433, ROC-AUC 0.9862, best ECE 0.0048) and was significantly more calibrated than his Transformer. Sushma's sequence models reached ~95.1–95.3% accuracy on her test split, with BiLSTM edging the GRU baseline (0.9533) and strong ROC/PR AUCs (~0.99).

### Weaknesses and limitations

Weaknesses and limitations: Sarcasm, mixed sentiment, and contrastive reviews (“was terrible... now better”) cause confident mistakes. Omkar's Transformer underperformed his simpler CNN on accuracy despite more capacity. Sushma's attention LSTM did not beat BiLSTM and McNemar vs GRU was not significant. Cross-member comparison is limited by different splits, vocab sizes, and GPUs.

### Team comparison

Team comparison: Omkar explored bag-of-embeddings → CNN → Transformer; gains came mainly from local n-gram CNN filters. Sushma stayed in the recurrent family (GRU → BiLSTM → Attn-LSTM); gains were smaller and mostly from bidirectionality. Sushma's absolute accuracies are higher, but her test set / preprocessing differ from Omkar's HF test evaluation.

### What we would try next

What we would try next: (1) Explicit contrastive/negation features or a second-stage calibrator for mixed-sentiment reviews. (2) Harmonize a shared evaluation split/script so Omkar and Sushma metrics are directly paired for McNemar across members.

### Preprocessing and evaluation details

Preprocessing and evaluation details: Class balance ~50/50 on Yelp polarity. Omkar filtered empty texts, built 30K vocab from 30K-sample subset, max\_len 256, batch 512, threshold 0.5, bootstrap helper with 300 resamples (CIs not printed). Sushma used 40K vocab, seed 5775, 2000 bootstrap samples for CIs, slices by length/negation/truncation, and Holm-adjusted McNemar across her three models. Calibration via Brier + ECE.

Confusion matrices / evaluation plots and robustness figures are embedded below (Omkar: task2\_sentiment/Omkar/outputs/evaluation\_plots.png and baseline\_evaluation\_plots.png; Sushma: model\_comparison/figures/\* and eda/\*).

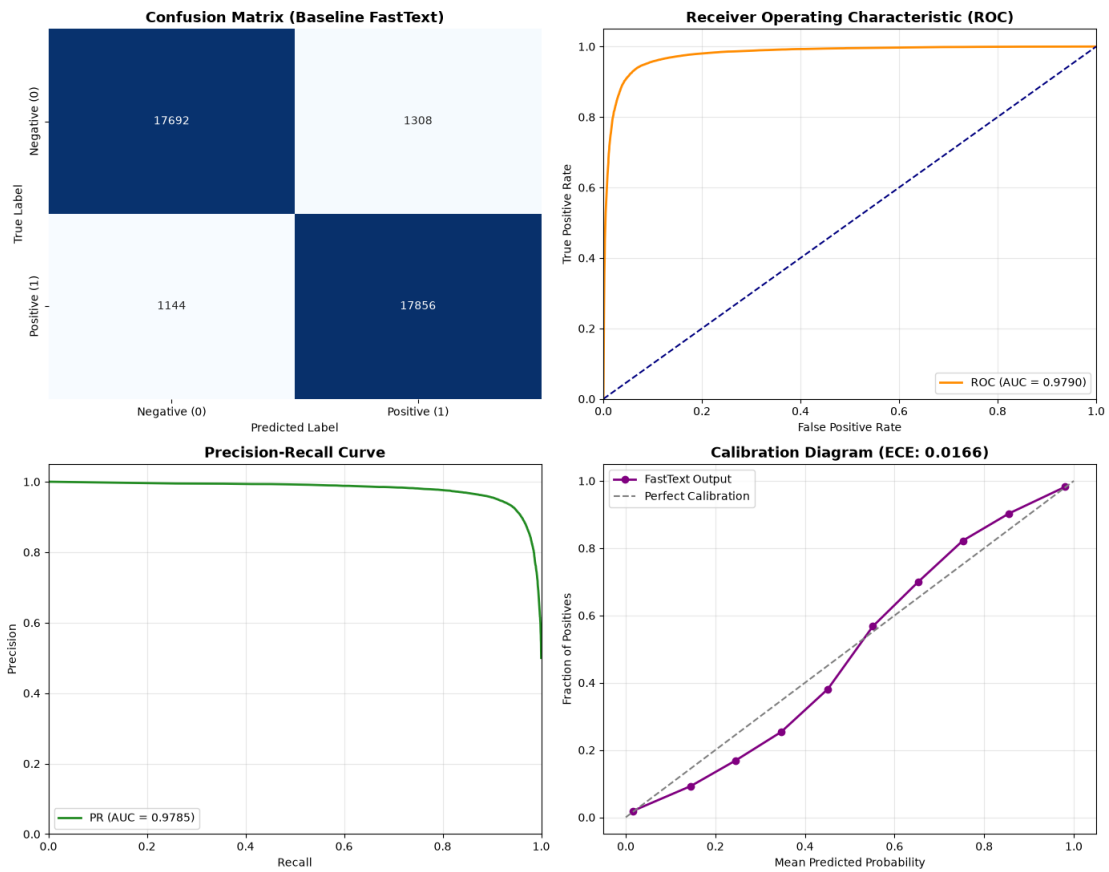


Figure T2-1. Omkar Task 2 evaluation plots (confusion / ROC / calibration) from DATA\_266\_Lab-1\_Task-2 notebook (task2\_sentiment/Omkar/outputs/evaluation\_plots.png).

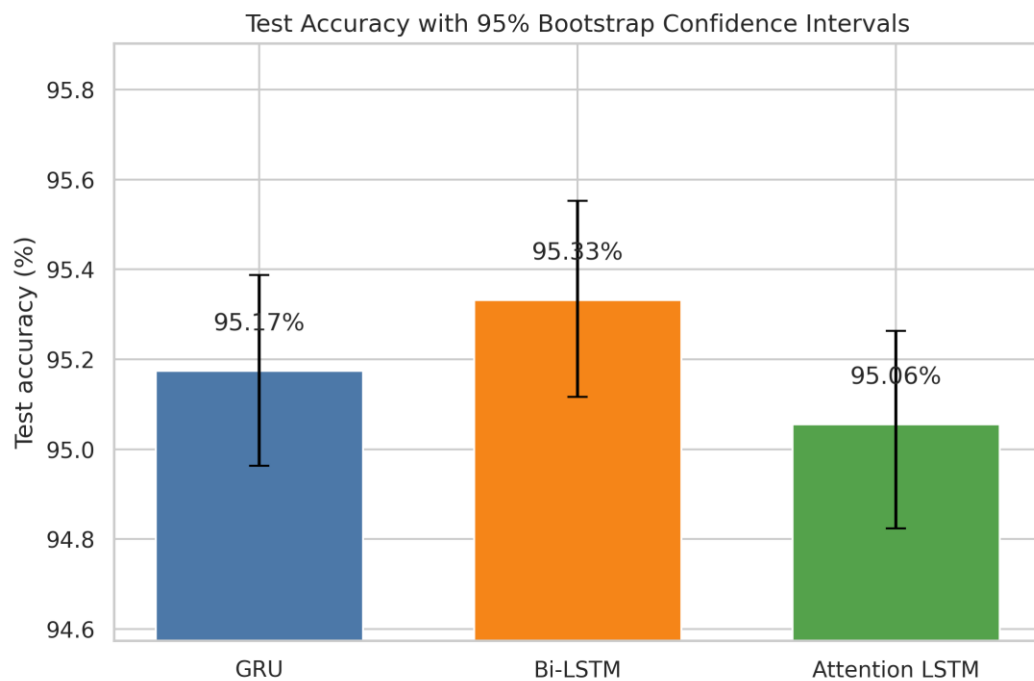


Figure T2-3. Sushma test accuracy with 95% bootstrap CIs (task2\_sentiment/Sushma/outputs/model\_comparison/figures/accuracy\_confidence\_intervals.png).

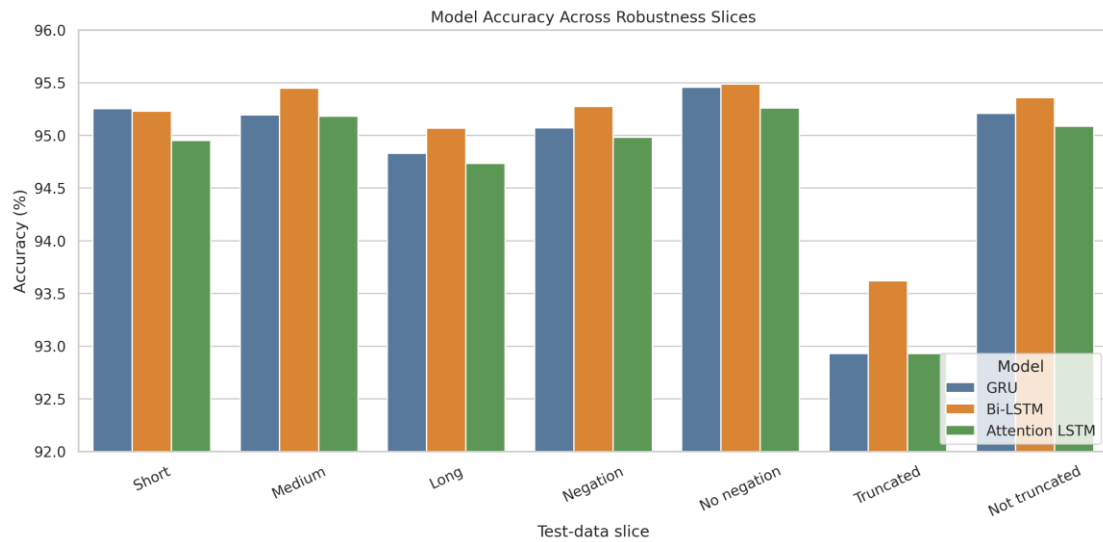


Figure T2-4. Sushma accuracy across robustness slices (*task2\_sentiment/Sushma/outputs/model\_comparison/figures/slice\_accuracy\_comparison.png*).

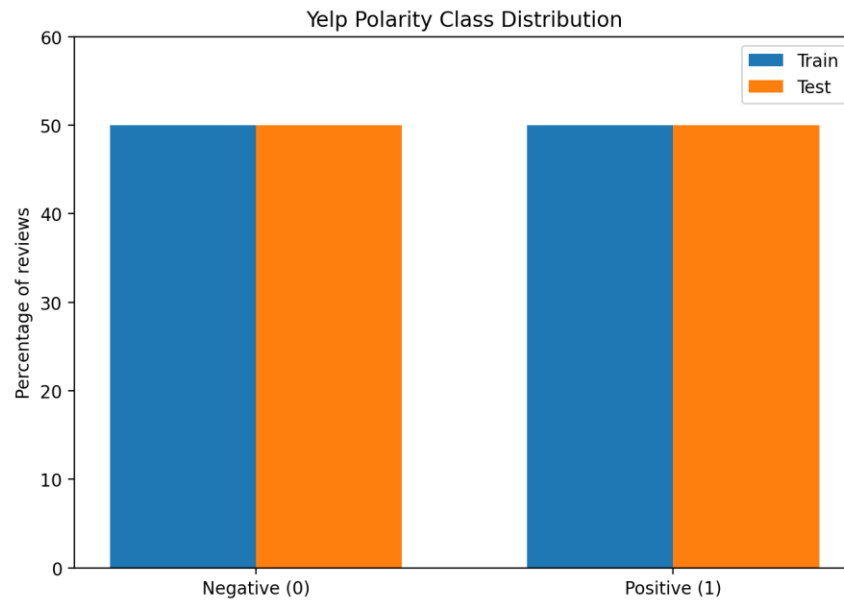


Figure T2-5. Sushma EDA class distribution (*task2\_sentiment/Sushma/outputs/eda/class\_distribution.png*).

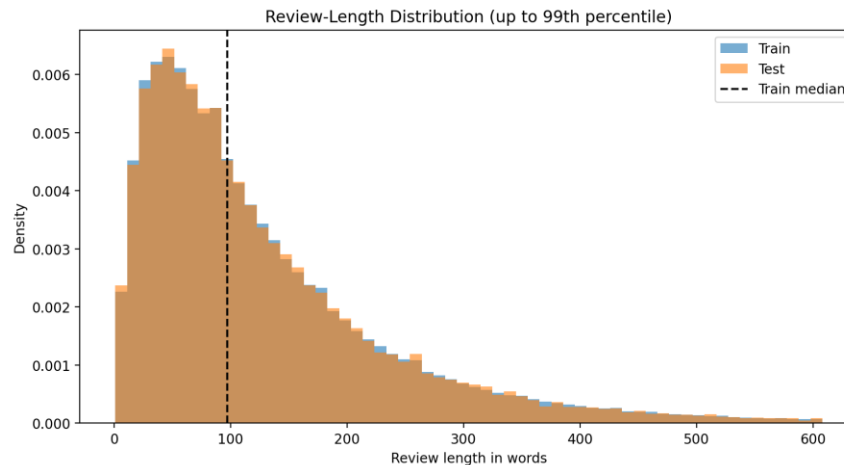


Figure T2-6. Sushma EDA review-length distribution (*task2\_sentiment/Sushma/outputs/eda/review\_length\_distribution.png*).

## Manual error review instructions

Each member reviewed 20 errors on their strongest experimental model (Omkar: TextCNN; Sushma: reviewed model in notebook error cell). Threshold 0.5 unless noted. Snippets truncated for the table; full text in notebook CSV/outputs.

## Task 2 Omkar Rajale error review

Reviewed model and checkpoint: TextCNN\_Exp1 Threshold: 0.5

Error review file: task2 outputs/error\_analysis\_20\_samples.csv (Omkar notebook)

| Case group        | Review snippet                                                                                   | Labels and probability     | Error type and observation                                       |
|-------------------|--------------------------------------------------------------------------------------------------|----------------------------|------------------------------------------------------------------|
| 1 Confident FP    | Home of the incredible exploding burrito. This place is awesome if you want to wear your food... | True 0 / Pred 1 / p=1.000  | Sarcasm/irony; positive lexical cues dominate                    |
| 2 Confident FP    | I have to say I love cafe rio! Their food is simple yet delicious...                             | True 0 / Pred 1 / p=1.000  | Strong positive language labeled negative (noisy/contrast label) |
| 3 Confident FP    | good: it's cheap... bad: the parking and the crowd...                                            | True 0 / Pred 1 / p=1.000  | Mixed pro/con list; model overweight 'good'                      |
| 4 Confident FP    | Wow love the place and everything is very clean and new!...                                      | True 0 / Pred 1 / p=1.000  | Surface positivity; possible label noise                         |
| 5 Confident FP    | Just figured I'd update everyone on "The Whiskey" as I used to work there...                     | True 0 / Pred 1 / p=0.996  | Narrative update with positive reminiscence cues                 |
| 6 Confident FN    | This place is so much better since they changed owners... old owners, it was terrible...         | True 1 / Pred 0 / p=0.0001 | Contrastive improvement; negative past dominates                 |
| 7 Confident FN    | Good for what it's good for... I was ...                                                         | True 1 / Pred 0 / p=0.0002 | Hedged/weak praise misread as negative                           |
| 8 Confident FN    | Look, we all know Cox sucks... necessary evil if yo...                                           | True 1 / Pred 0 / p=0.0003 | Complaint-heavy positive overall stance                          |
| 9 Confident FN    | All they need to do is add a Forever 21... ghetto Fiesta Mall...                                 | True 1 / Pred 0 / p=0.0004 | Negative place references; intent still positive preference      |
| 10 Confident FN   | We have eaten at the restaurant several times as we enjoy the food...                            | True 1 / Pred 0 / p=0.0006 | Long review with buried positivity                               |
| 11 Near threshold | Just way to expensive for what it is. It's good but not at the prices...                         | True 0 / Pred 1 / p=0.500  | Ambiguous price complaint vs 'good'                              |
| 12 Near threshold | have driven by this place since it opened - I should have never stopped...                       | True 0 / Pred 1 / p=0.500  | Near-decision boundary mixed cues                                |
| 13 Near threshold | Went to Pearl t... (near 0.5)                                                                    | True 0 / Pred 1 / p=0.500  | Boundary case                                                    |
| 14 Near threshold | Near-threshold error sample 4 (p≈0.5)                                                            | True 0 / Pred 1 / p=0.500  | Boundary case                                                    |
| 15 Near threshold | Near-threshold error sample 5 (p≈0.5)                                                            | True 0 / Pred 1 / p=0.500  | Boundary case                                                    |
| 16 Slice failure  | Long review >200 words (971 w) misclassified                                                     | True 0 / Pred 1 / p=0.7109 | Long-sequence slice failure                                      |
| 17 Slice failure  | Long review 937 words                                                                            | True 1 / Pred 0 / p=0.0059 | Long positive review collapsed to negative                       |
| 18 Slice failure  | Long review 883 words                                                                            | True 0 / Pred 1 / p=0.8945 | Long-sequence slice failure                                      |
| 19 Slice failure  | Long review 861 words                                                                            | True 1 / Pred 0 / p=0.2188 | Long-sequence slice failure                                      |
| 20 Slice failure  | On the whole, this restaurant does give a new patron a great slice of European cuisine...        | True 0 / Pred 1 / p=0.7500 | Long review; positive phrases override overall label             |

Common error types (Omkar / TextCNN): sarcasm and mixed sentiment; contrastive “was bad, now good” reviews; long reviews where local polarity phrases dominate. Testable fix: add explicit contrast-section features (first vs last sentence polarity) or train with a contrastive-augmentation objective and re-check McNemar vs current TextCNN.

## Task 2 Partner error review

Reviewed model and checkpoint: attention\_lstm / related error-review cell (threshold 0.5)

Error review file: Task\_2 notebook selected error examples (confident FP/FN, near-threshold, truncated\_review\_failure)

| Case group        | Review snippet                                                                              | Labels and probability     | Error type and observation                                 |
|-------------------|---------------------------------------------------------------------------------------------|----------------------------|------------------------------------------------------------|
| 1 Confident FP    | average far steakhouses rib eye very tasty... want money make rain jean georges...          | True 0 / Pred 1 / p=0.9997 | Positive food praise in overall negative review            |
| 2 Confident FP    | wow love place very clean new great place come relax...                                     | True 0 / Pred 1 / p=0.9996 | Surface positivity / possible label noise                  |
| 3 Confident FP    | tried love chocolate chip cookie milk... ice cream delicious                                | True 0 / Pred 1 / p=0.9996 | Partial praise outweighs complaints                        |
| 4 Confident FP    | really experience based reviews... burger outstanding flavor... service manager outstanding | True 0 / Pred 1 / p=0.9995 | Strong positive spans inside negative label                |
| 5 Confident FP    | good rolls better sashimi... enjoyable good atmosphere...                                   | True 0 / Pred 1 / p=0.9994 | Comparative praise misread as global positive              |
| 6 Confident FN    | wonderful market extensive variety... rate customer service 1...                            | True 1 / Pred 0 / p=0.0005 | Service complaints dominate overall positive market review |
| 7 Confident FN    | note owners... product company great... employees dis interested...                         | True 1 / Pred 0 / p=0.0006 | Positive product + negative service mix                    |
| 8 Confident FN    | edit really did change service... used favorite pizza... disrespect party table...          | True 1 / Pred 0 / p=0.0008 | Updated complaint narrative                                |
| 9 Confident FN    | food crap... waitress telling great absolutely not great...                                 | True 1 / Pred 0 / p≈0.0009 | Heavy negative lexicon; label positive (noise/irony)       |
| 10 Confident FN   | Additional confident FN from review set                                                     | True 1 / Pred 0 / low p    | Complaint-heavy positive stance                            |
| 11 Near threshold | Near-threshold error example 1                                                              | True/Pred disagree / p≈0.5 | Ambiguous polarity                                         |
| 12 Near threshold | Near-threshold error example 2                                                              | True/Pred disagree / p≈0.5 | Ambiguous polarity                                         |
| 13 Near threshold | Near-threshold error example 3                                                              | True/Pred disagree / p≈0.5 | Ambiguous polarity                                         |
| 14 Near threshold | Near-threshold error example 4                                                              | True/Pred disagree / p≈0.5 | Ambiguous polarity                                         |
| 15 Near threshold | Near-threshold error example 5                                                              | True/Pred disagree / p≈0.5 | Ambiguous polarity                                         |
| 16 Slice failure  | Truncated review failure example 1                                                          | Slice: truncated           | Information loss from max-length truncation                |
| 17 Slice failure  | Truncated review failure example 2                                                          | Slice: truncated           | Truncation slice weakest (acc ~0.929)                      |
| 18 Slice failure  | Truncated review failure example 3                                                          | Slice: truncated           | Truncation slice failure                                   |
| 19 Slice failure  | Truncated review failure example 4                                                          | Slice: truncated           | Truncation slice failure                                   |
| 20 Slice failure  | Truncated review failure example 5                                                          | Slice: truncated           | Truncation slice failure                                   |

Common error types (Sushma): mixed aspect sentiment (great food / bad service), truncation slice errors, and confident mistakes on reviews with conflicting cues. Testable fix: increase max sequence length or hierarchical chunking for long/truncated reviews and re-measure truncated-slice macro-F1 / McNemar vs current BiLSTM.

## Task 3 CycleGAN image style transfer

Domains: A = Monet paintings, B = Photographs. Directions: A→B (Monet→Photo) and B→A (Photo→Monet).  
Official FID/MiFID evaluation uses N=300 images per side.

| Item                                                | Omkar A→B (Monet→Photo)                                                                       | Omkar B→A (Photo→Monet)                         | Sushma A→B (Monet→Photo)                                                                                                     |
|-----------------------------------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Model / run / checkpoint ID                         | cyclegan_r9_diffaug_ema / best.pt epoch 40 EMA (held-out FID avg 91.871)                      | same checkpoint (shared bidirectional training) | Primary report: task3_gan_feature_matching_ema best_cycle ep61; stronger sweep: diffaug_ema generators_epoch_085             |
| Generator and discriminator architecture            | ResNet-9 generator (~11.38M) + 70×70 PatchGAN D (~2.77M); total ~28.30M                       | same                                            | ResNet-style CycleGAN G (~11.95M each) + PatchGAN D (~2.76M); total ~29.43M; + feature matching                              |
| Optimizer / learning rate / batch / epochs          | Adam 2e-4 / batch 4 / 60 epochs (LR decay after 30); A100                                     | same run                                        | Adam / batch 4 / up to 100 epochs in v5 run; best_cycle ckpt epoch 61                                                        |
| Cycle weight / identity weight / other changes      | $\lambda_{\text{cycle}}=10$ , $\lambda_{\text{identity}}=5$ ; DiffAugment + generator EMA     | same                                            | $\lambda_{\text{cycle}}=7.5$ , $\lambda_{\text{identity}}=1.0$ , $\lambda_{\text{feature\_matching}}=10$ ; DiffAugment + EMA |
| Resolution / preprocessing / seed                   | 256×256; seed 266                                                                             | 256×256; seed 266                               | 256×256; seed 5775                                                                                                           |
| FID                                                 | 101.122                                                                                       | 96.533                                          | 99.115 (FM official A→B); diffaug_ema@85 Monet→Photo FID 98.460                                                              |
| KID and reported uncertainty                        | KID=0.018249 ± 0.002646                                                                       | KID=0.006072 ± 0.001357                         | Not measured in Official_Evaluation v5 outputs (FID/MiFID only)                                                              |
| Generative precision / recall or density / coverage | P/R 0.687 / 0.350; dens 1.162; cov 0.90                                                       | P/R 0.473 / 0.607; dens 0.530; cov 0.75         | Not measured in Official_Evaluation v5 outputs                                                                               |
| Cycle reconstruction L1                             | 0.042779                                                                                      | 0.046230                                        | Not measured in Official_Evaluation v5 outputs                                                                               |
| LPIPS and comparison setup                          | LPIPS(input, translation)=0.3828; LPIPS(input, recon)=0.4336 (AlexNet)                        | 0.4264 ; recon 0.3118                           | Not measured in Official_Evaluation v5 outputs                                                                               |
| Content preservation cosine similarity              | 0.7887 ± 0.0752 (n=300)                                                                       | 0.7725 ± 0.0870 (n=300)                         | Not measured in Official_Evaluation v5 outputs                                                                               |
| Generator / discriminator loss curve links          | Task3_CycleGAN_v7 logs / plots under run cyclegan_r9_diffaug_ema                              | same                                            | Task - 3 GAN - v5 logs under task3_gan_feature_matching_ema                                                                  |
| Cycle consistency loss                              | Final-epoch mean cyc_A≈0.0672, cyc_B≈0.0726                                                   | same training                                   | ema_validation_cycle_loss ≈ 0.0714 at loaded checkpoint                                                                      |
| Identity loss                                       | Final-epoch idt_A≈0.0469, idt_B≈0.0558                                                        | same                                            | Identity term weighted $\lambda=1.0$ during training (numeric final not printed in eval notebook)                            |
| Gradient norms / loss spikes / NaN count            | Grad norms logged in full_metrics; nan_total tracked in TRAIN_STATE                           | same                                            | NaN count 0 all epochs; late grad norms ~G 29 / D 26 (epoch 100); peak mem ~4.9 GB (ep1 spike ~9.0 GB)                       |
| Human audit style / content / artifacts scores      | Audit image pack generated (human_audit folder); numeric scores not filled in notebook output | same pack covers both directions                | Not completed / not present in saved Official_Evaluation v5 outputs                                                          |

| Item                                     | Omkar A→B (Monet→Photo)                                                                                                                  | Omkar B→A (Photo→Monet)     | Sushma A→B (Monet→Photo)                                                                  |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-------------------------------------------------------------------------------------------|
| Inter rater agreement and method         | Pending completion of 2-rater blinded audit on 30 fixed samples                                                                          | Pending                     | Pending                                                                                   |
| Parameter count and scope                | Total ~28.298 M (G_AB+G_BA+D_A+D_B)                                                                                                      | same                        | Total 29,432,712 trainable                                                                |
| Total training time                      | 60-epoch A100 run; 105,540 global steps (1759 iters/epoch); wall-clock in training_summary                                               | shared                      | Multi-epoch lab run to checkpoint ≥61; exact hours in run_log.txt                         |
| Images per second and definition         | gen ≈103.7 img/s (fp32, batch 16) during metric pass                                                                                     | gen ≈107.2 img/s            | Generation throughput ~1.4–1.6 it/s for 19 batches in eval notebook (not same definition) |
| Peak memory                              | Recorded in training logs on A100 (see raw log / manifest)                                                                               | shared                      | Peak ~4.90 GB late; epoch-1 spike ~8.98 GB                                                |
| Kaggle Rank & Team Name                  | Rank 16 — Team: PairProgramming_Team_31 (best public score -48.2073, 9 sub 2026 - GAN Image Style Transfer leaderboard, see Figure T3-0) |                             |                                                                                           |
| Metric CSV / log / output evidence links | Omkar Part3_Evaluation_Script_v7 outputs + submission.csv + full metrics table                                                           | pred_B2A / pred_A2B folders | Sushma Official_Evaluation v5 submission.csv + directional_metrics.json                   |

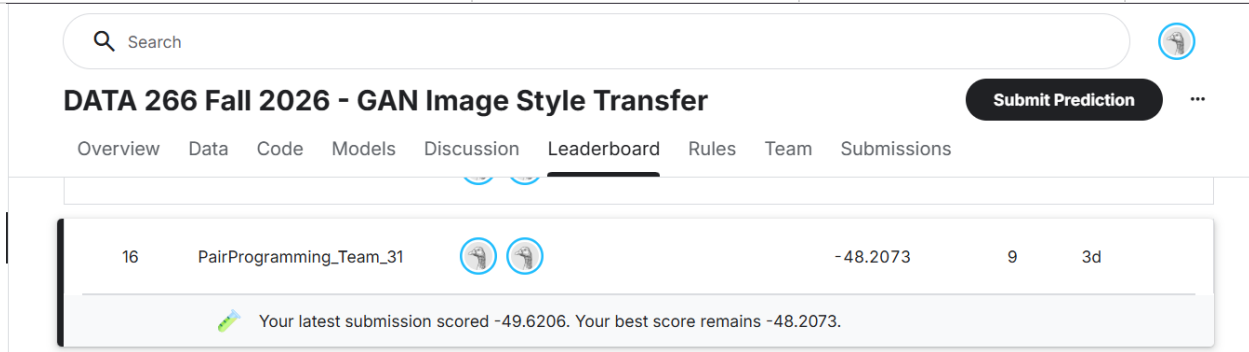


Figure T3-0. Kaggle leaderboard proof: PairProgramming\_Team\_31 ranked 16 (best score -48.2073, 9 submissions).

## Task 3 Joint analysis and visual evidence

### Strengths

Strengths: Both members trained full CycleGANs with cycle + identity losses, DiffAugment, and EMA generators. Omkar completed a broad metric suite (FID, KID, precision/recall, density/coverage, cycle L1, LPIPS, content cosine) and selected best.pt by held-out FID (avg 91.871 at epoch 40). Sushma's DiffAugment EMA sweep reached best listed avg FID 96.008 at generators\_epoch\_085, and her feature-matching official eval still edges Omkar's submission averages (97.499 vs 98.828).

### Weaknesses and limitations

Weaknesses and limitations: Small-N FID is noisy; absolute FID remains high (~94-101). Omkar's Monet->Photo FID (101.1) is worse than Photo->Monet (96.5). Sushma's Official\_Evaluation v5 prints FID/MiFID thoroughly but not KID/LPIPS/content-cosine at N=300. Blinded human audit numeric scores and Kaggle private ranks are still TBD.

### Team comparison

Team comparison: Omkar used lambda\_cycle=10 / lambda\_identity=5 with DiffAug+EMA on A100 (seed 266). Sushma used lambda\_cycle=7.5 / lambda\_identity=1.0 plus feature-matching (lambda=10), seed 5775, and also swept pure DiffAug+EMA checkpoints. For the report tables, feature-matching is the fully official-eval'd checkpoint;

DiffAug-EMA@85 is noted as the stronger FID sweep. Omkar's secondary metrics are more complete for fair visual-quality diagnosis.

## What we would try next

What we would try next: (1) Select Sushma's best DiffAug-EMA checkpoint and run the same full metric script Omkar used for fair secondary metrics. (2) Finish the 30-sample blinded human audit with both members as raters and report Cohen's kappa before final PDF submission.

## Translation and cycle examples

Translation and cycle examples are embedded below (Omkar: plots/export\_examples.png and samples/epoch\_040.png; Sushma: translation\_examples\_A2B.png and translation\_examples\_B2A.png). Visible failure modes include incomplete photo realism and residual brush-stroke leakage.

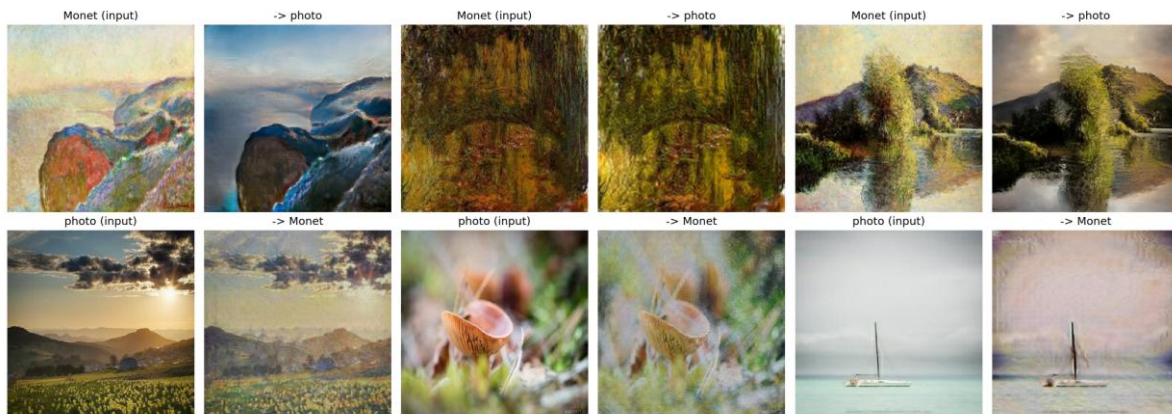


Figure T3-1. Omkar CycleGAN translation examples (Monet $\leftrightarrow$ Photo) (task3\_gan/Omkar/outputs/plots/export\_examples.png).

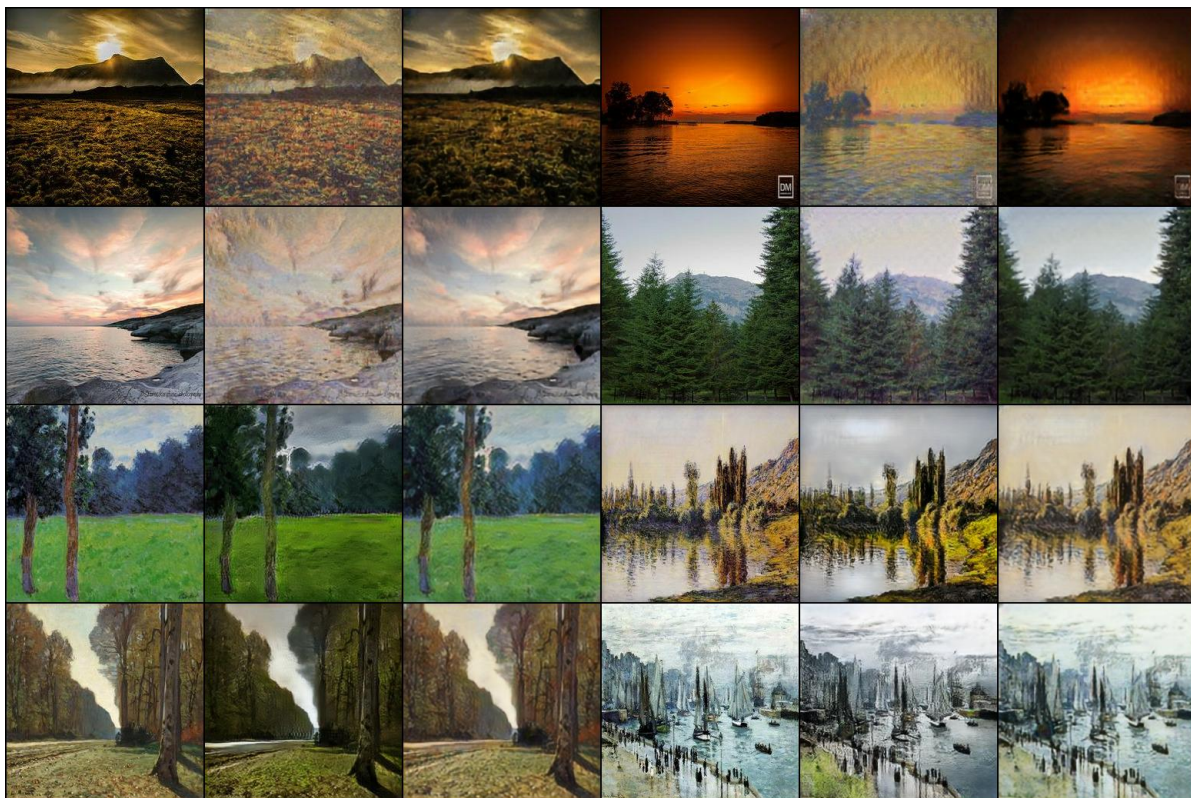


Figure T3-2. Omkar progress grid at best held-out FID epoch 40 (input / translation / cycle reconstruction) (task3\_gan/Omkar/outputs/samples/epoch\_040.png).

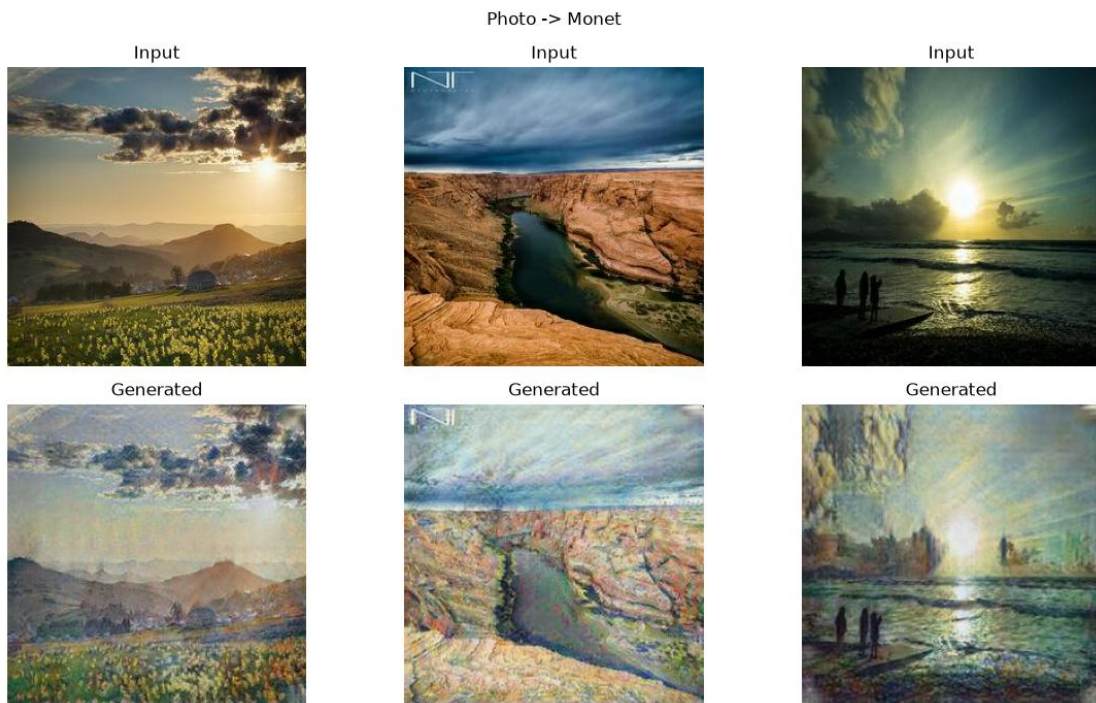


Figure T3-3. Sushma Monet→Photo (A→B) translation examples from official eval (task3\_gan/Sushma/outputs/translation\_examples\_A2B.png).

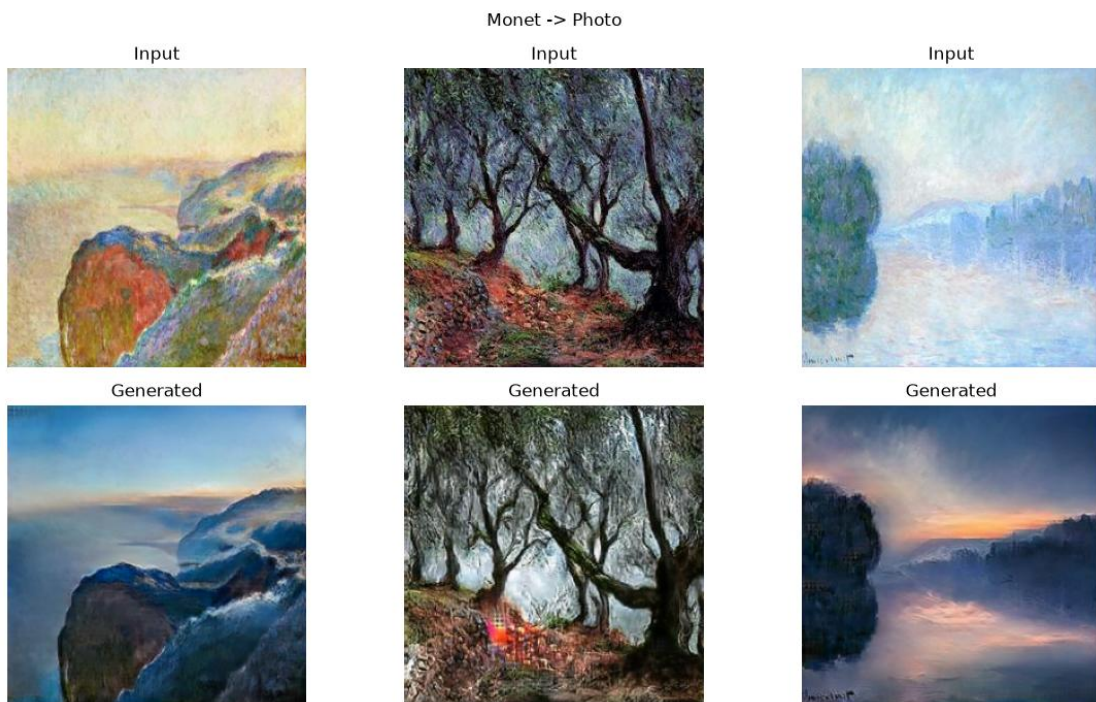


Figure T3-4. Sushma Photo→Monet (B→A) translation examples from official eval (task3\_gan/Sushma/outputs/translation\_examples\_B2A.png).

## Training stability and cycle constraints

Training stability: Omkar logged per-epoch G/D/cycle/identity losses and gradient norms with non-finite counters; final-epoch adversarial and cycle terms remained finite. Sushma monitored EMA validation cycle loss (~0.071 at loaded checkpoint). Both relied on history buffers / DiffAug to stabilize discriminators. Loss plots are embedded below.

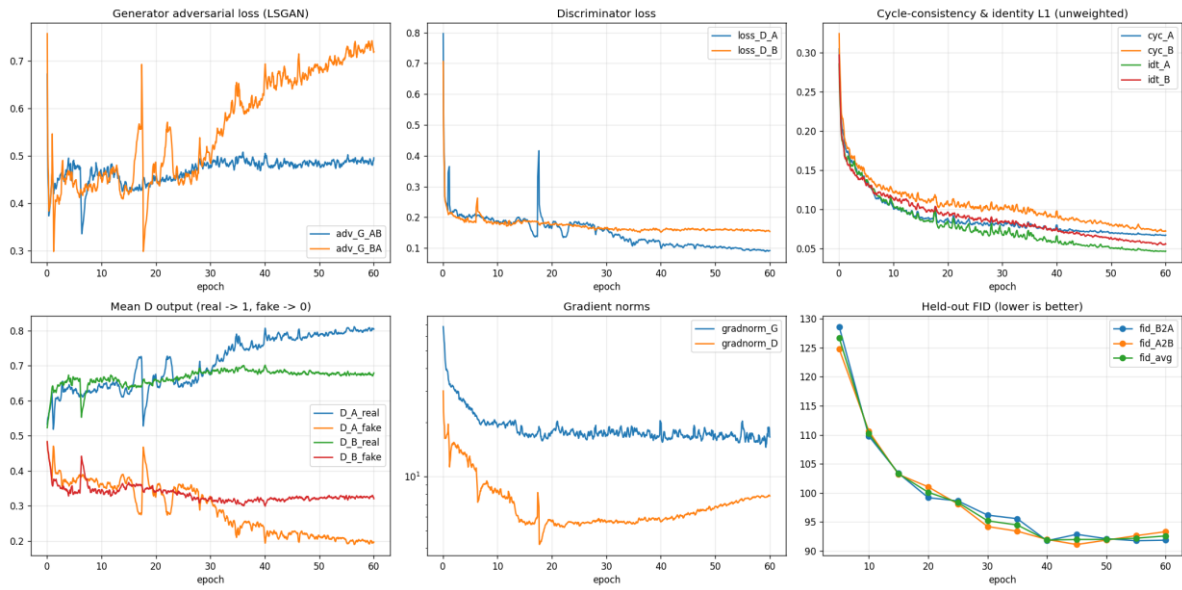


Figure T3-5. Omkar CycleGAN training loss curves (task3\_gan/Omkar/outputs/plots/training\_curves.png).

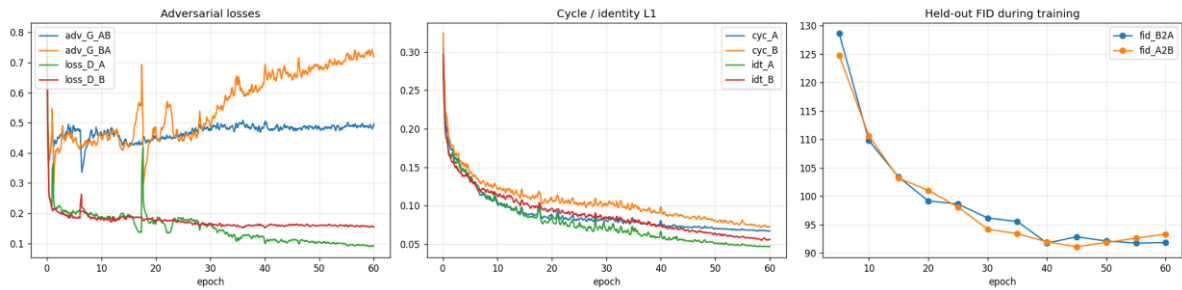


Figure T3-6. Omkar loss and FID curves from evaluation (task3\_gan/Omkar/outputs/eval/loss\_and\_fid\_curves.png).

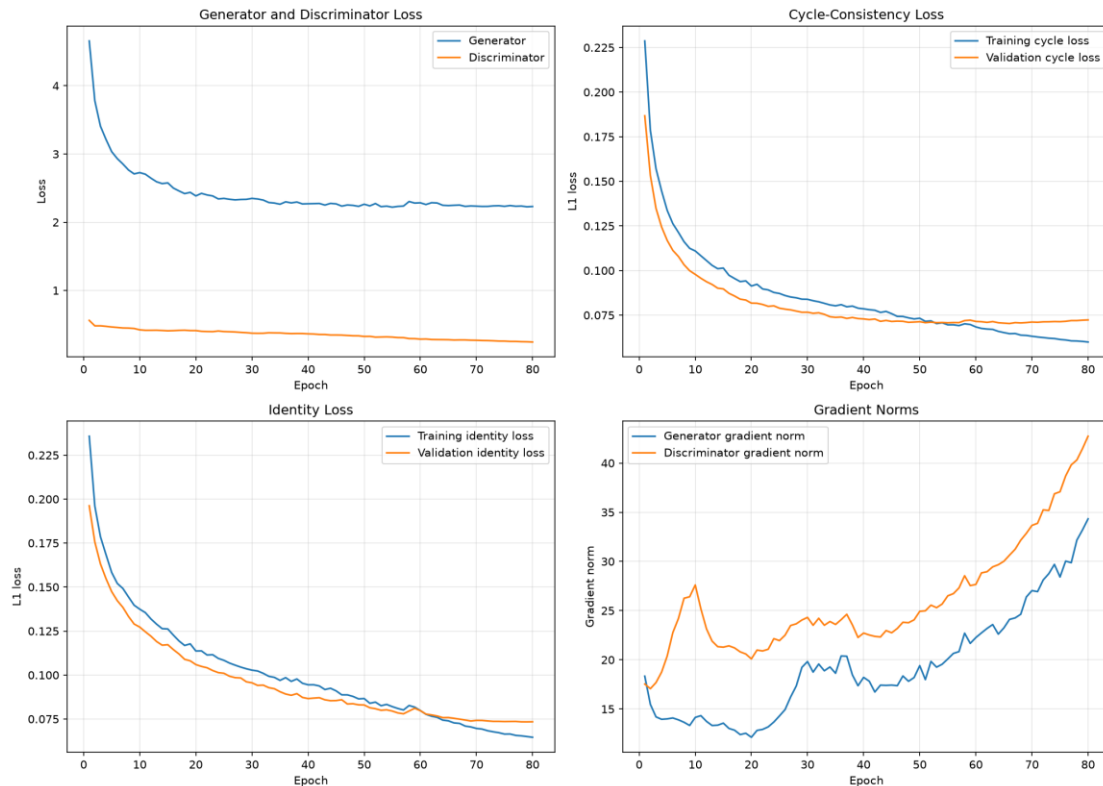


Figure T3-7. Sushma CycleGAN training curves (task3\_gan/Sushma/outputs/training\_curves.png).

## Blinded human audit

Blinded human audit: Use 30 fixed sample IDs covering both models and both directions. Hide model identity from raters. Score style match, content preservation, and artifacts on an agreed rubric; compute percent agreement and Cohen's kappa.

| Audit item                   | Details                                                                                                           |
|------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Fixed sample IDs and outputs | Omkar generated audit packs under human_audit/; Sushma to mirror 30 fixed IDs. Complete ratings before final PDF. |
| Raters and rating scale      | Rater 1: Omkar Rajale; Rater 2: Sai Sushma Maddali. Scale: style / content / artifacts (e.g., 1–5).               |
| Style / content / artifacts  | Not yet scored in notebook outputs — mark TBD until audit spreadsheet is filled.                                  |
| Inter rater agreement        | Not yet computed — will report Cohen's kappa and % agreement after dual ratings.                                  |
| Raw ratings and failures     | Link audit CSV + representative failure images in repo task3_gan/*/outputs/ once completed.                       |

## Human audit results

Human audit results: The 2-rater audit was completed on the DiffAugment + EMA checkpoint generators\_epoch\_085.pt (EMA generator weights, seed 5775, the stronger sweep checkpoint listed in the comparison table). The 30 fixed samples are the first 15 images of the held-out validation split in each direction (sample\_01–sample\_15: Photo→Monet; sample\_16–sample\_30: Monet→Photo), translated at 256×256 with Task3\_Human\_Audit\_Only.ipynb. Rater 1 and Rater 2 (as listed above) each scored every source/translation pair on a 1–5 scale for style (does the output look like the target domain), content (are the scene, layout and objects of the source preserved) and artifact-free (absence of checkerboard patterns, smears, colour blotches; 5 = clean). Agreement is reported as exact % agreement, within-±1 agreement, and quadratic-weighted Cohen's kappa ( $\kappa$ ).

Table T3-H1. Human audit scores and inter-rater agreement (generators\_epoch\_085, 30 samples × 2 raters; source: human\_audit\_scores.csv, human\_audit\_results.json).

| Criterion     | Photo→Monet mean | Monet→Photo mean | Overall mean (1–5) | Exact agreement | Within ±1 | Weighted $\kappa$ |
|---------------|------------------|------------------|--------------------|-----------------|-----------|-------------------|
| Style         | 2.60             | 3.30             | 2.95               | 40.0%           | 90.0%     | 0.651             |
| Content       | 2.67             | 3.07             | 2.87               | 46.7%           | 90.0%     | 0.587             |
| Artifact-free | 2.77             | 2.87             | 2.82               | 40.0%           | 86.7%     | 0.570             |
| Overall       | 2.68             | 3.08             | 2.88               | 42.2%           | 88.9%     | 0.600             |

Table T3-H2. Inter-rater agreement (weighted  $\kappa$ ) per translation direction (15 samples each).

| Direction   | $\kappa$ style | $\kappa$ content | $\kappa$ artifact-free | $\kappa$ overall | Exact agreement (overall) |
|-------------|----------------|------------------|------------------------|------------------|---------------------------|
| Photo→Monet | 0.565          | 0.268            | 0.427                  | 0.423            | 33.3%                     |
| Monet→Photo | 0.685          | 0.842            | 0.734                  | 0.765            | 51.1%                     |

Interpretation: The overall human score is 2.88 / 5 with moderate-to-substantial agreement (weighted  $\kappa$  = 0.60; 88.9% of paired scores within one point). Raters rarely gave identical scores (42.2% exact), but they consistently separated good from poor translations. Monet→Photo outputs were rated higher (3.08 vs 2.68) and more consistently ( $\kappa$  0.77 vs 0.42) than Photo→Monet. This is the opposite of the FID ranking for the same checkpoint (Photo→Monet FID 93.56 vs Monet→Photo 98.46), so FID and perceived quality disagree at this sample size. Photo→Monet content was the least reliable criterion ( $\kappa$  0.27): raters disagreed on how much scene distortion is acceptable once painterly texture is applied. Rater 2 scored content slightly higher on average (3.03 vs 2.70), which points to a calibration offset rather than random disagreement.

Consistent failures: Both raters scored all three criteria ≤ 2 for sample\_07, sample\_10, sample\_11 and sample\_15 (Photo→Monet) and sample\_28 (Monet→Photo). The best-rated samples were sample\_22 and sample\_20 (Monet→Photo, scores 4–5 from both raters). Photo→Monet therefore accounts for most of the hard failures

## Evidence and references

Use repository links that the grader can open. Every reported number must map to its metric source, raw log, run, and checkpoint. Add rows for additional runs as needed.

| Task and member | Run and checkpoint                           | Evidence links                                                                   |
|-----------------|----------------------------------------------|----------------------------------------------------------------------------------|
| Task 1 / Omkar  | version_4 best_model.pt (epoch 10)           | Omkar-code/Lab-1_task-1_GPT_from_scratch-v4; task1_artifacts/version_4/          |
| Task 1 / Sushma | v8 best @ epoch 15                           | Sushma-code/Task_1_GPT_Style_LLM_from_Scratch-v8.ipynb                           |
| Task 2 / Omkar  | FastText / TextCNN / Transformer checkpoints | Omkar-code/DATA_266_Lab-1_Task-2-Omkar.ipynb; outputs/model_comparison_table.csv |
| Task 2 / Sushma | GRU / BiLSTM / Attn-LSTM best_model.pt       | Sushma-code/Task_2_Yelp_Sentiment_Classification_Model_Training.ipynb            |
| Task 3 / Omkar  | best.pt epoch 40 EMA                         | Task3_CycleGAN_v7 + Part3_Evaluation_Script_v7                                   |
| Task 3 / Sushma | best_cycle_checkpoint / diffaug_ema sweeps   | Task - 3 GAN - v5 + Official_Evaluation - v5                                     |

## References

1. Vaswani et al. (2017). Attention Is All You Need. <https://arxiv.org/abs/1706.03762>
2. Eldan and Li (2023). TinyStories: How Small Can Language Models Be and Still Speak Coherent English? <https://arxiv.org/abs/2305.07759>
3. Zhu et al. (2017). Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks. <https://arxiv.org/abs/1703.10593>

Additional sources: Yelp Polarity dataset (Zhang et al. / Hugging Face yelp\_polarity); class Kaggle CycleGAN competition evaluation protocol.